MUTHU VARSHITH

Bengaluru, India · muthuvarshith290@gmail.com · +91 9100073813
GitHub: MuthuVarshith · LinkedIn: Muthu Varshith · Portfolio: muthu-varshith-portfolio.vercel.app

PROFESSIONAL SUMMARY

Python Developer and AI/ML Engineer building LLM agents, RAG systems and deep learning services with FastAPI, PostgreSQL, Docker and CI. Measures design choices on labelled benchmarks with significance tests, keeps humans in control of AI decisions, and backs projects with 1,000+ automated tests.

TECHNICAL SKILLS

Programming: Python, SQL, JavaScript, HTML, CSS
Generative AI: LLM Agents, Tool Calling, LangGraph, RAG, Hybrid Retrieval (BM25, RRF, Rerankers), Embeddings, Prompt Engineering, RAGAS, LLM-as-Judge
AI / ML: PyTorch, Hugging Face Transformers, timm, scikit-learn, LoRA (PEFT), Computer Vision, NLP, Model Evaluation (AUC, McNemar, Bootstrap)
Backend: FastAPI, Flask, Pydantic, REST APIs, Server-Sent Events, Webhooks, SQLAlchemy, Alembic
Databases: PostgreSQL + pgvector, Qdrant, SQLite
MLOps / DevOps: Docker, Docker Compose, GitHub Actions, MLflow, ONNX, Langfuse, Terraform, AWS (EC2, IAM, SSM)
Data & Tools: NumPy, Pandas, Matplotlib, Seaborn, pytest, ruff, mypy, Git

PROJECTS

Cerberus — Agentic Software Repair with a Verification Gate *Aug 2026 – Present*

- Built an agentic bug-repair system where a tool-calling LLM agent (Anthropic, Gemini, OpenRouter) proposes patches and a deterministic gate (RED, GREEN, regression and scope checks in an offline Docker sandbox) decides; 10 of 10 valid real-model repairs were admitted with 0 false admissions on hidden tests.
- Designed code retrieval with Tree-sitter chunking, BM25, Qdrant vector search and RRF fusion, raising bug-localization Recall@10 from 0.49 to 0.86 on 120 SWE-bench Verified instances (McNemar $p < 10^{-5}$).
- Shipped a FastAPI GitHub App with issue-triggered repairs, LangGraph orchestration with checkpointed resume and human approval before any draft PR; the gate cut wrong approvals from 65% to 36% while admitting 7 of 7 correct fixes.

HireAI — AI Recruiting Platform with RAG and a Human-Approved Agent *Jul 2026 – Present*

- Built a FastAPI and PostgreSQL + pgvector recruiting platform with a LangGraph agent whose shortlists are saved only after human approval, and cited RAG answers streamed over Server-Sent Events.
- Implemented hybrid retrieval (vector, BM25, RRF, cross-encoder reranking) and measured it on a 30-question labelled benchmark under a pre-committed protocol; kept vector search (MRR@10 0.833) as no addition showed a reliable gain.
- Secured with argon2id passwords, hashed server-side sessions and rate-limited login, with each prompt-injection guard tested against an unguarded twin; delivered Docker Compose and GitHub Actions CI with 639 passing tests.

FusionTrace — Privacy-First Deepfake Detection Service *Mar 2025 – Present*

- Rebuilt a FastAPI and PyTorch service that scores images, audio and video locally using swappable detectors fused into a weighted, explained risk score, with API keys and human review labels.
- Evaluated on out-of-distribution data and replaced an image model that failed on external sets with a ViT-S detector (98.9% on AI-generated faces; not built for face swaps); Wav2Vec2 audio scored 98.3% on a 4,002-clip held-out set and 87% on In-the-Wild.
- Added LoRA fine-tuning (221K trainable parameters) with leakage-aware splits, MLflow tracking and registry, ONNX export and INT8 quantisation; withheld fine-tune results after a shortcut check exposed a dataset artifact.

WORK EXPERIENCE

Python Full Stack Development Intern | **KodNest** *Feb 2026 – May 2026*

- Developed a Python-based inventory, sales and purchase management system covering stock management, multi-item invoicing, returns and payments, with role-based access control for admin, sales and inventory teams.
- Implemented stock and billing logic in atomic database transactions to enforce non-negative inventory, and validated the business rules with 59 automated tests covering overselling, returns and access control.

Artificial Intelligence Intern | **SmartEd Solutions** *Jun 2025 – Oct 2025*

- Trained and compared Logistic Regression and SVM classifiers in scikit-learn on the 569-sample Breast Cancer Wisconsin (Diagnostic) dataset after exploratory data analysis and feature standardization.
- Achieved 97.4% held-out accuracy with Logistic Regression versus 95.6% with SVM on an 80/20 split, evaluated with confusion matrices and ROC curves.

EDUCATION

Visvesvaraya Technological University — B.E. in Computer Science and Business Systems *2022 – 2026*

CERTIFICATIONS

Oracle (2025): OCI Generative AI Professional · OCI Data Science Professional · Fusion AI Agent Studio Associate
AWS: Foundations of Prompt Engineering · **Nasscom Foundation:** Machine Learning & Image Processing